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Division: A2

Subject: Programming with Java

ASSIGNMENT No : 10

TITLE :Write a Java program to demonstrate the use of abstract classes and abstract methods

Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

Theory

Abstract classes provide partial implementation of a program and cannot be instantiated directly.

They may contain both abstract and concrete methods. Abstract methods declare functionality that

must be implemented by subclasses. Abstraction hides implementation details while exposing only

essential functionality. This improves code flexibility and maintainability in object-oriented

programming.

Outcome

Conclusion

These programs demonstrate **abstraction** by using abstract classes and methods, allowing different subclasses to provide their own implementation of the required functionality.

Exercise

1.Create an abstract Payment class and implement Credit Card and UPI payment classes.

```
import java.util.Scanner;

abstract class Payment {
    abstract void pay(double amount);
}

class CreditCard extends Payment {
    void pay(double amount) {
        System.out.println("Paid ₹" + amount + " using Credit Card.");
    }
}

class UPI extends Payment {
    void pay(double amount) {
        System.out.println("Paid ₹" + amount + " using UPI.");
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Amount: ");
        double amount = sc.nextDouble();

        Payment p1 = new CreditCard();
```

```
        Payment p2 = new UPI();

        p1.pay(amount);
        p2.pay(amount);
    }
}
```

2. Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

```
import java.util.Scanner;

abstract class FoodOrder {
    abstract void calculateBill(double amount);
}

class DineInOrder extends FoodOrder {
    void calculateBill(double amount) {
        double total = amount + 50; // Service charge
        System.out.println("Dine-In Bill = " + total);
    }
}

class TakeAwayOrder extends FoodOrder {
    void calculateBill(double amount) {
        double total = amount + 20; // Packing charge
        System.out.println("Take Away Bill = " + total);
    }
}
```

```java
public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Food Amount: ");
        double amount = sc.nextDouble();

        FoodOrder d = new DineInOrder();
        FoodOrder t = new TakeAwayOrder();

        d.calculateBill(amount);
        t.calculateBill(amount);
    }
}
```